

Data Labeling Guidelines - VR Testing

This document contains the steps, examples, and information regarding the labeling process for the skeleton joint data regarding the VR testing project.

Scope

This project's scope is the PhantomLimb Game suite, specifically Bubble Burst.

The data collected from this application represents the X, Y, Z coordinates of the virtual objects present in the virtual scene.

Goal

The goal is to identify issues that are present at the start of the game or to detect any possible problem that can appear during the gameplay.

For this, we are going to use our own subjective reasoning to label the extracted data to identify any issues that can impact the user's experience in playing the game. In order to do so, we identified two distinct categories for the discovered issues.

Terminology

Skeleton configuration

- **Clarification:** The skeleton is displayed by 24 joints and the bones that connect them, corresponding to the user's skeleton. Mathematically evaluating if the given skeleton is in a correct human configuration is very hard. In this case, we are assessing if the displayed skeleton looks correct in our subjective opinion or if there are any issues that might affect the gameplay of the user.

Spatial aspects

- **Clarification:** Spatial aspects relate to:
 - the configuration of the user's skeleton at the given moment
 - the display of the environment and static objects in the scene
 - the relation between the skeleton and any static objects in the scene through gameplay, where the skeleton is treated as a static object.
- It assesses if both the skeleton and environment are correctly configured, positioned, and visually appropriate for gameplay at any given moment.
- This impacts initial impressions and gameplay setup.
- **Key Question:** *Does the skeleton's configuration and environment appearance meet the standards for an acceptable gameplay experience (when viewed at a single point in time)?*
- **Key question:** *Do we need to restart/reconfigure the game? (if the user or environment configuration affects the game)*

Temporal aspects

- **Clarification:** Temporal aspects involve:
 - the configuration of the user's skeleton, considering its display over time,
 - the skeleton's continuous interaction with the dynamic objects in the scene (e.g. bubbles), where the skeleton is treated as a dynamic object
- It evaluates if the skeleton is successfully displayed and can interact with other dynamic virtual objects throughout gameplay.
- This affects ongoing usability, playability, and user experience.
- **Key Question:** *Does the skeleton's configuration or performance in interacting with game elements impair the user's experience (during the game session)?*
- **Key Question:** *Do we need to stop the game? (if the issue is too big to ignore)*

Labeling process

For the labeling process, we will use a Python script shared in a Google Colab notebook.

We will label 3 independent elements:

- the spatial relation of the user's skeleton and the floor
- the temporal relation of the user's skeleton and the bubbles
- the image of the user playing the game

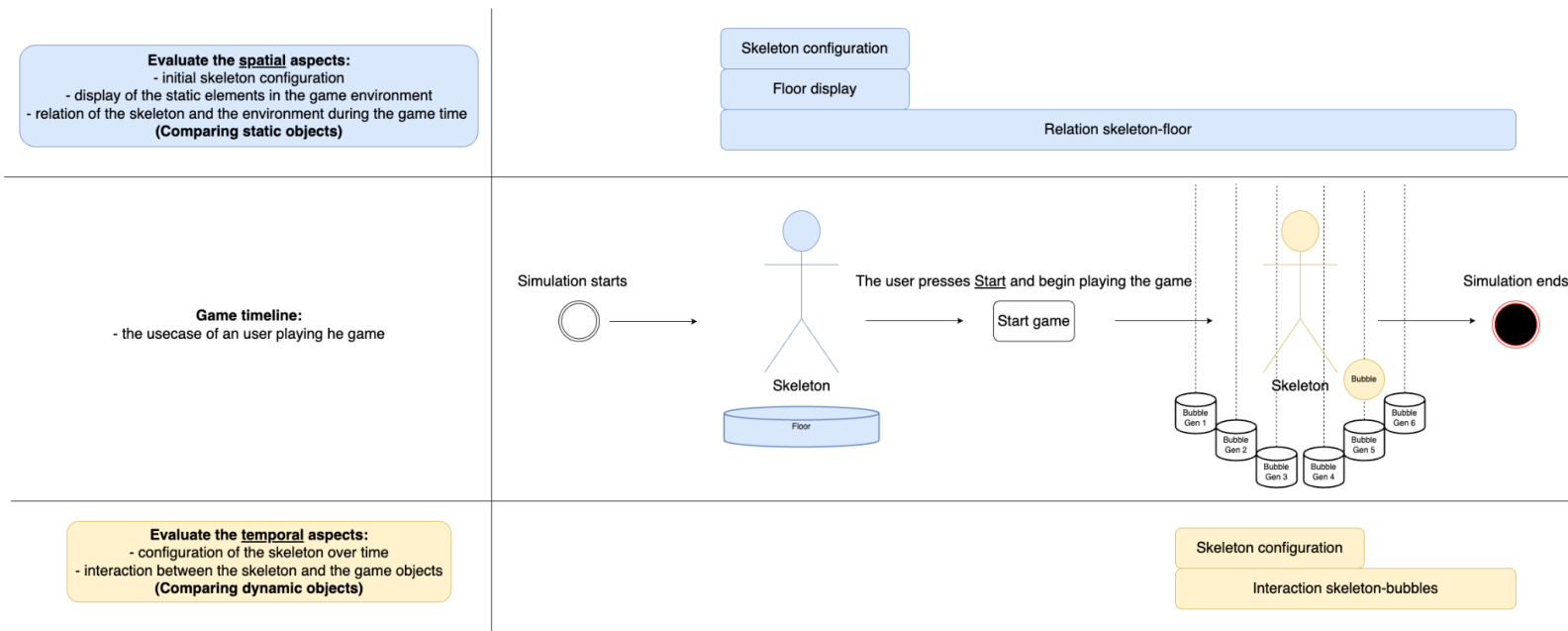
Each of these elements will be labeled with:

- 0 (false): if we identify any of the spatial or temporal issues that can impact the game
- 1 (true): if there are no issues **OR** if the identified issues are not big enough to make an impact

The data points have different IDs and will be displayed in batches.

Each student will be assigned a section of the data that he needs to label.

The number of displayed data points will be left to the choice of each student. We recommend trying batches of 20-50 data points at a time.



Important considerations

*When evaluating the temporal aspects, please take into consideration that we are evaluating the relation of the skeleton with the bubbles over time, for this we will consider 2 previous data points and 2 post-data points in our judgment (when present). Please keep this in mind when starting a new labeling batch. (remember the previous data points labels or start the new labeling batch with the last 2 data points from the previous batch).

*Please take into consideration that the game has strict rules on how to use it, but the users can still do unexpected actions. If these actions are different from the given instructions for using the game it does not mean necessarily that we need to label it negatively (0) or stop the game. In this case, we need to use our subjective opinion if the unexpected action is an issue that we can overlook (labeled with 1) or an issue that we think will affect the gameplay (labeled with 0).

Conflict resolution

Any data point that is labeled differently by 2 students will be evaluated again by a third student and the original-taskeds students to eliminate any inconsistencies.

Examples

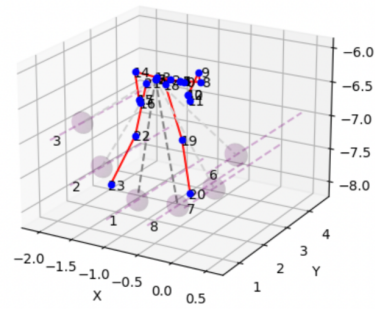
Temporal:

Good (1) :

- Correct display and distance:

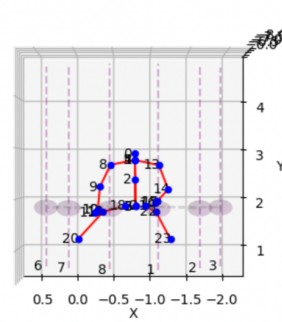


View 1 - Human Skeleton 5448

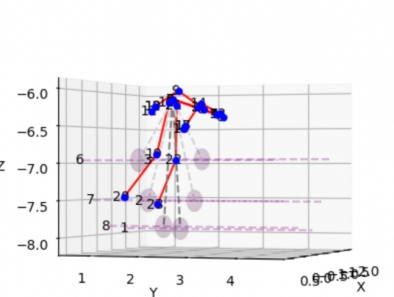


Interaction: ○ 0
● 1

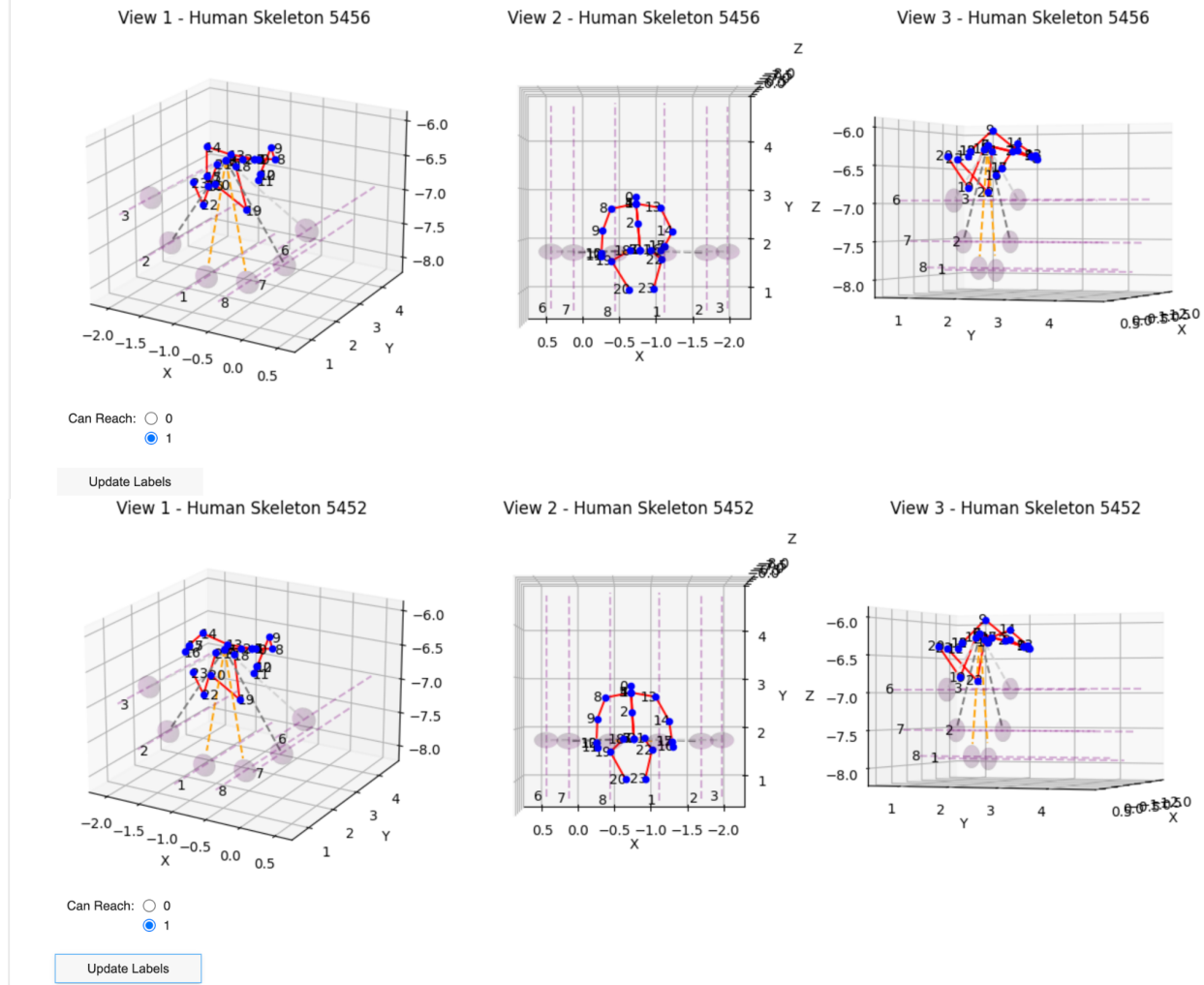
View 2 - Human Skeleton 5448



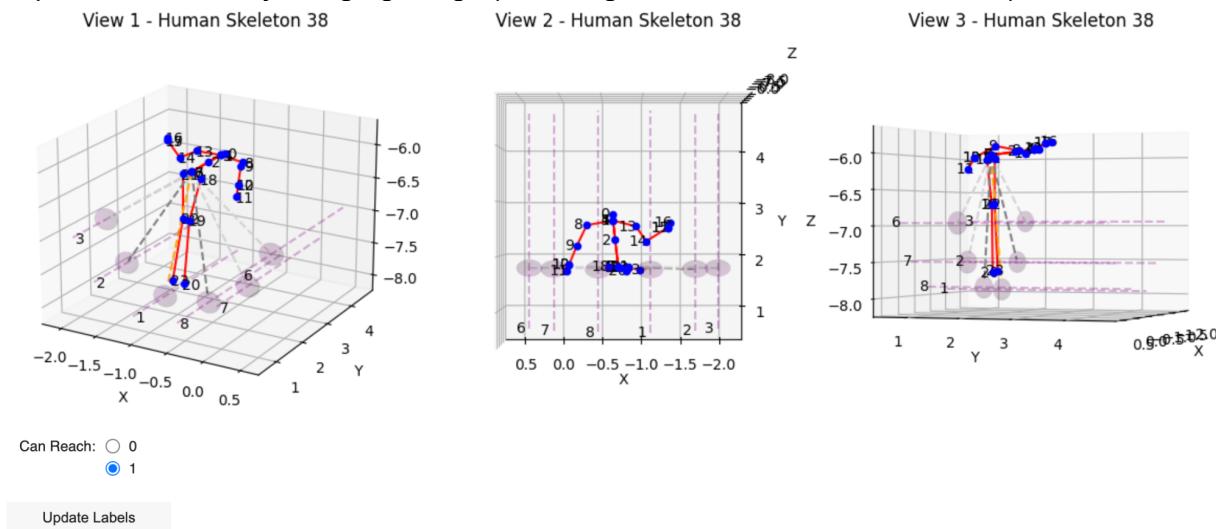
View 3 - Human Skeleton 5448



- Small leniency with the bubble distance:

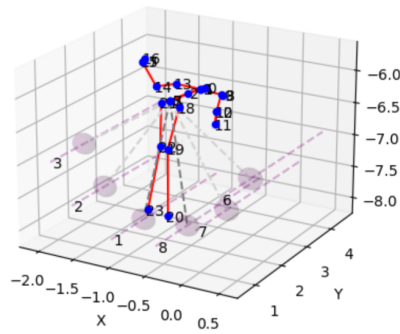


- Accepted inconsistency in leg/right legs (if one leg reaches, the other should too):



- Good example:

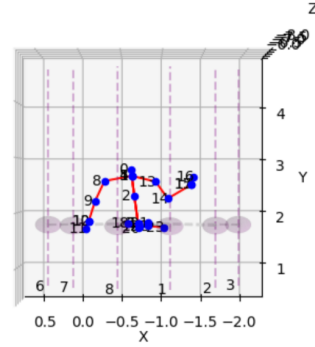
View 1 - Human Skeleton 42



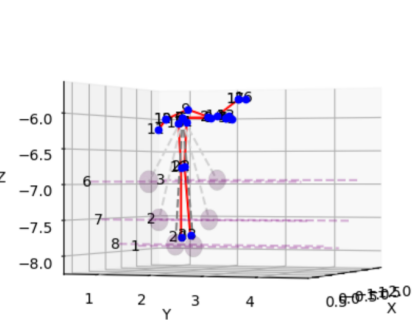
Can Reach: ○ 0
● 1

Update Labels

View 2 - Human Skeleton 42

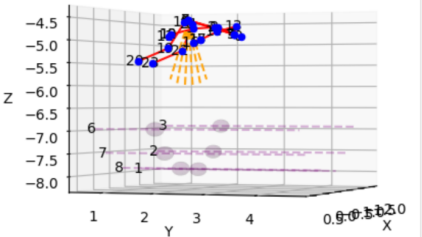
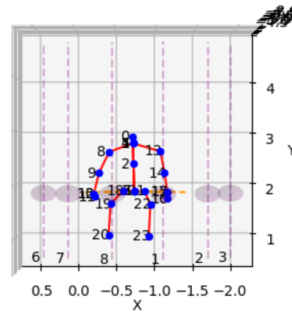
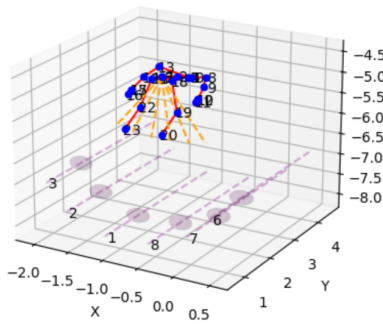


View 3 - Human Skeleton 42



Bad (0):

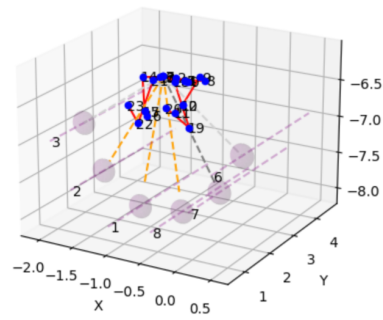
- The skeleton is far away from the camera:



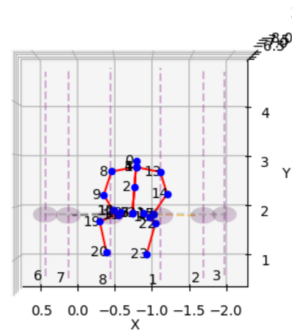
- The skeleton cannot reach 3/6 bubbles, we consider this a clear problem:

...

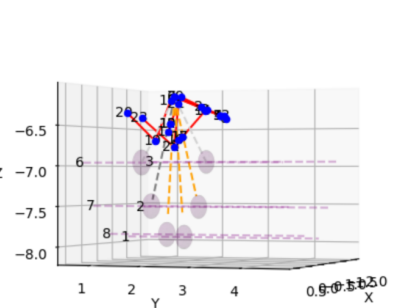
View 1 - Human Skeleton 5443



View 2 - Human Skeleton 5443

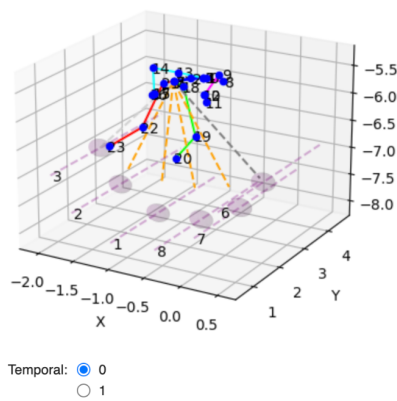


View 3 - Human Skeleton 5443

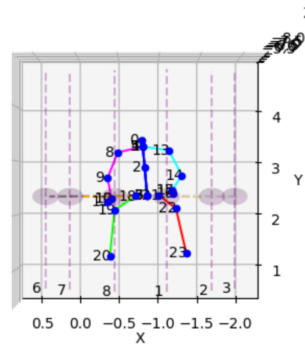


- This is considered

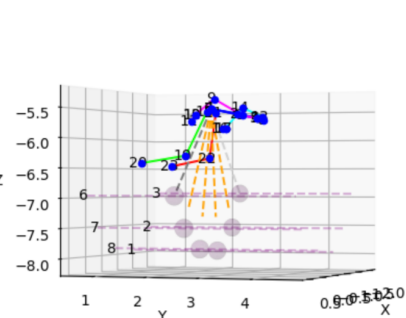
Angle 1 - Human Skeleton 755



Angle 2 - Human Skeleton 755



Angle 3 - Human Skeleton 755

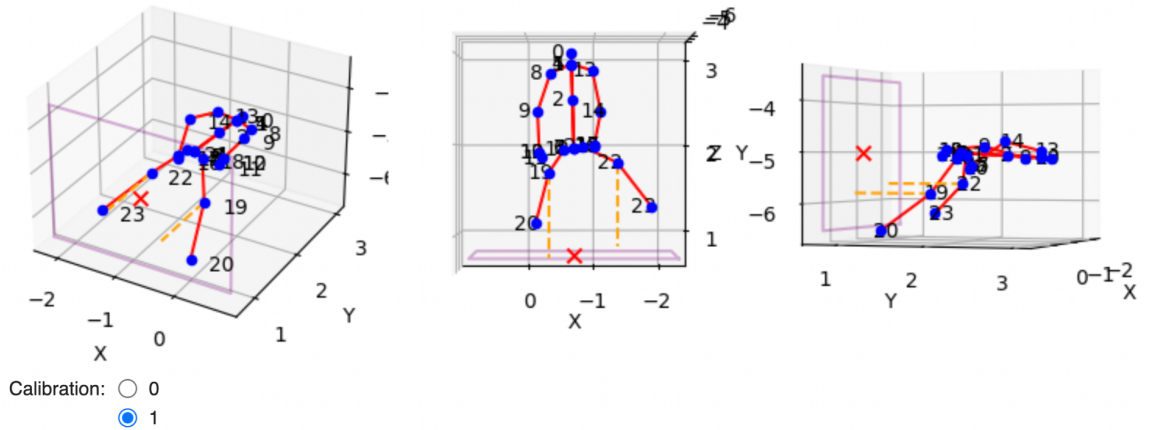


Spatial:

Good (1):

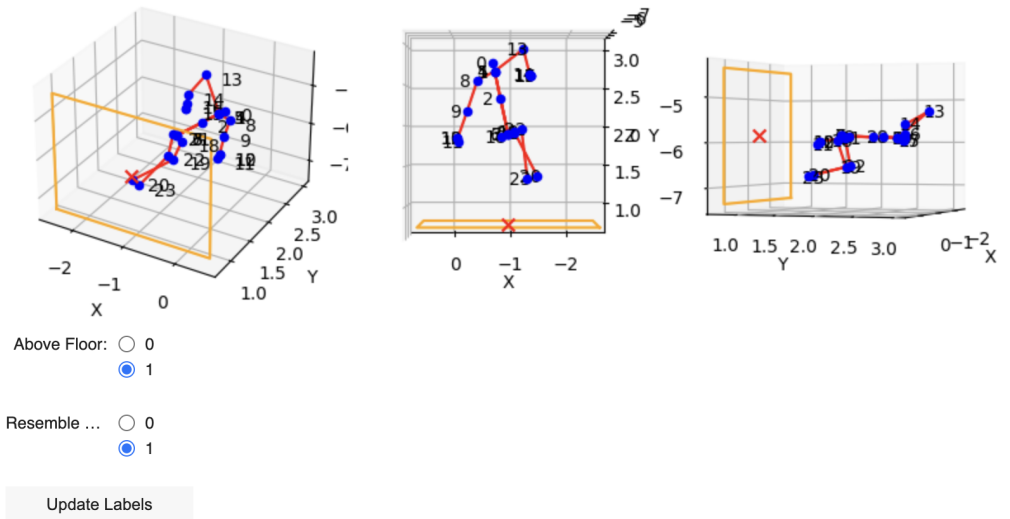
- Small leniency with the floor:

View 1 - Human Skeleton 5771 View 2 - Human Skeleton 5771 View 3 - Human Skeleton 5771



- Unexpected configuration , but accepted:

View 1 - Human Skeleton 24 View 2 - Human Skeleton 24 View 3 - Human Skeleton 24

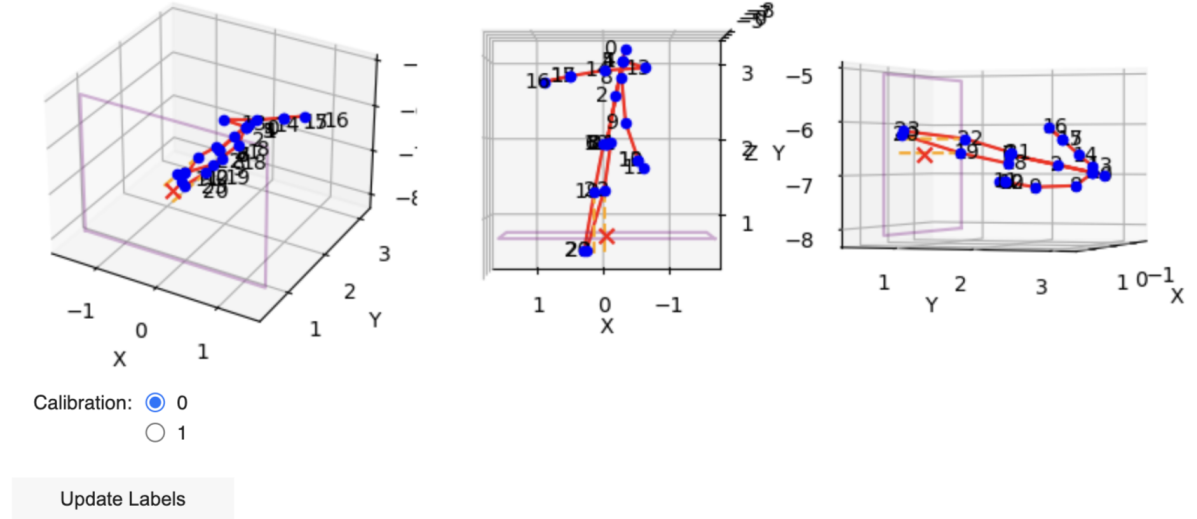


-

Bad (0):

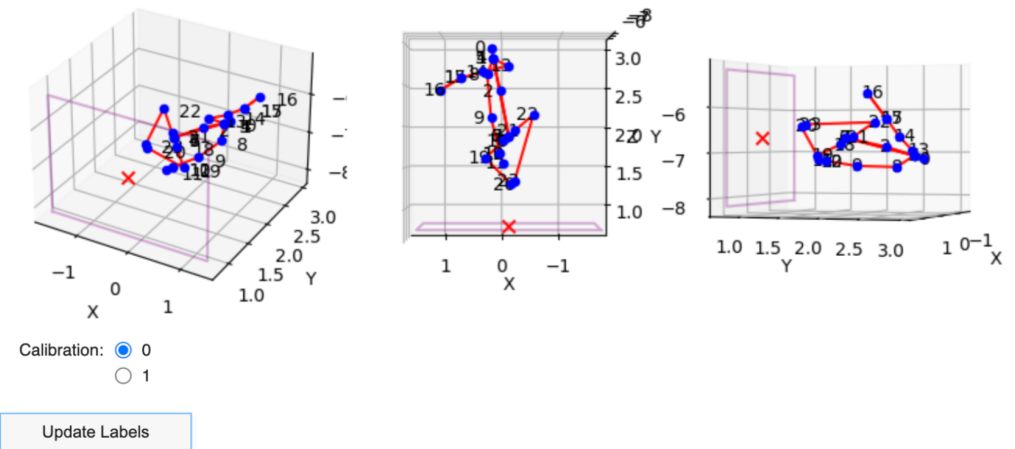
- Legs under floor:

View 1 - Human Skeleton 331 View 2 - Human Skeleton 331 View 3 - Human Skeleton 331

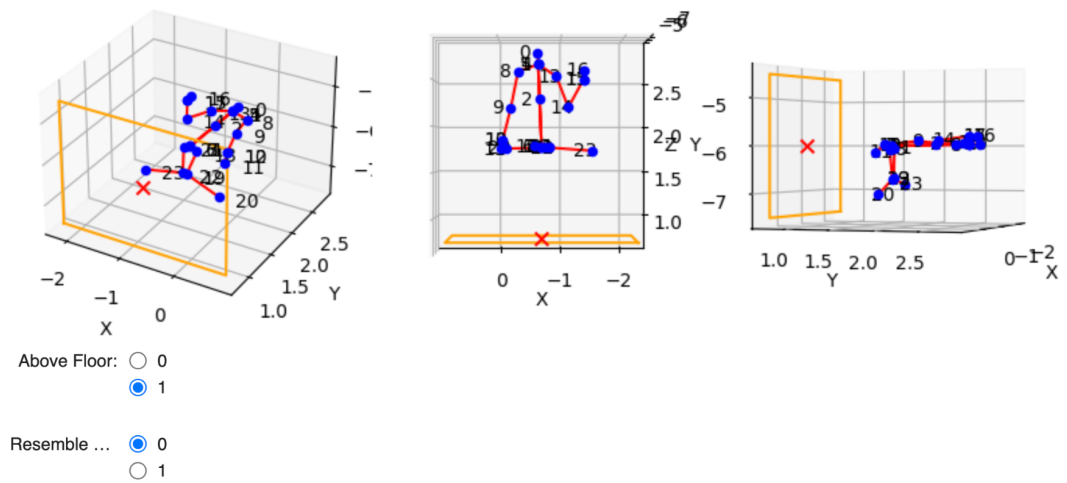


- Configuration issues (weird or unexpected postures or shapes):

View 1 - Human Skeleton 332 View 2 - Human Skeleton 332 View 3 - Human Skeleton 332

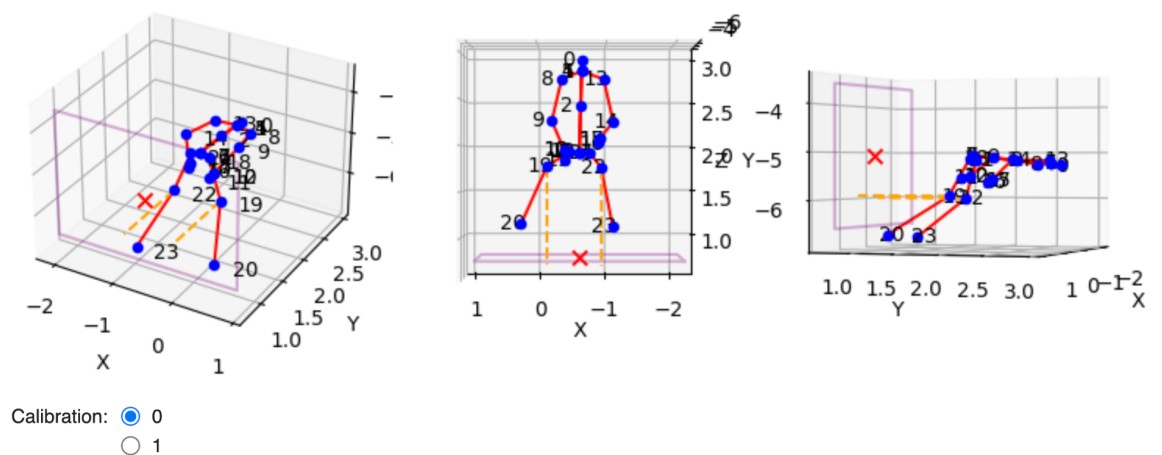


View 1 - Human Skeleton 79 View 2 - Human Skeleton 79 View 3 - Human Skeleton 79

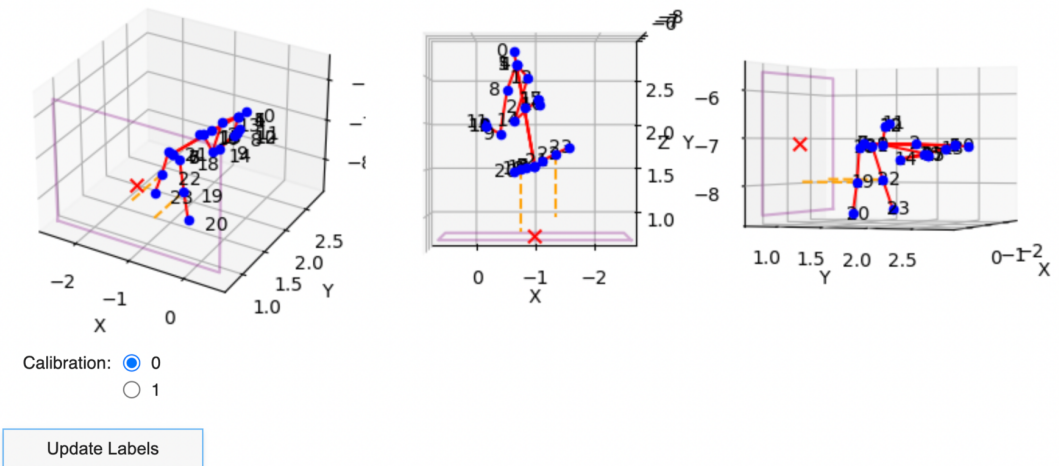


- Leg projection for sitting pose (orange dotted line) under the floor:

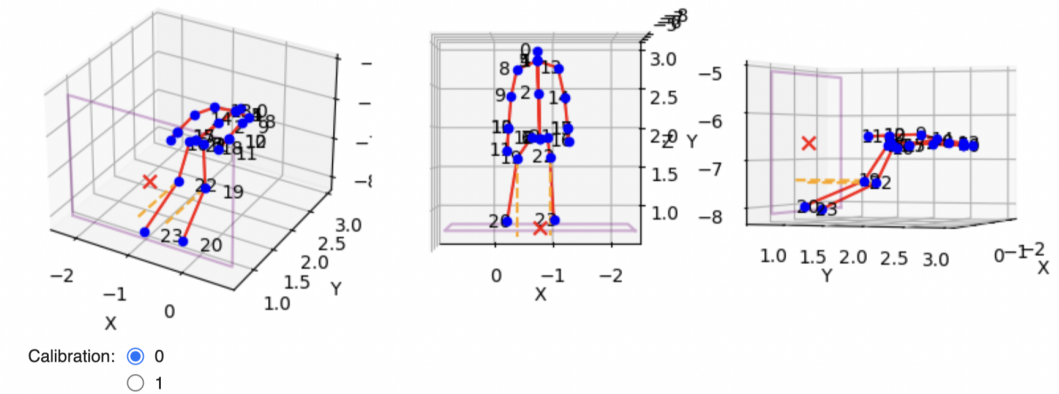
View 1 - Human Skeleton 5694 View 2 - Human Skeleton 5694 View 3 - Human Skeleton 5694



View 1 - Human Skeleton 663 View 2 - Human Skeleton 663_Z View 3 - Human Skeleton 663

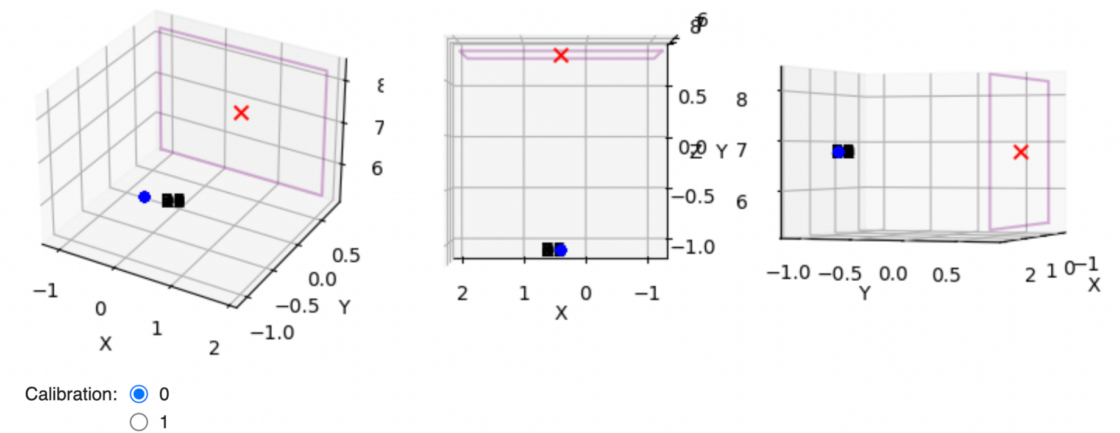


View 1 - Human Skeleton 5361 View 2 - Human Skeleton 5361_Z View 3 - Human Skeleton 5361

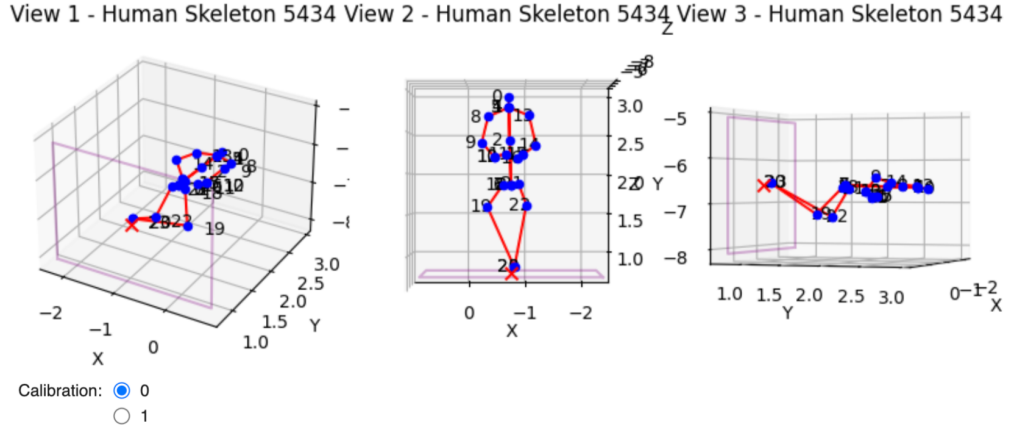


- The skeleton did not load properly:

View 1 - Human Skeleton 581 View 2 - Human Skeleton 581_Z View 3 - Human Skeleton 581



- Overlapping joints (e.g. in this case):



Summary

To label the **Spatial** aspects,

- Please look for:
 - The skeleton is correctly displayed (**NO** missing limbs or weird positions, **NO** missing person in the image)
 - The skeleton is facing the camera
 - The skeleton is positioned correctly above the floor, and **allow** a small leniency
- Please **disregard**:
 - The skeleton is not in the middle of the room

To label the **Temporal** aspects,

- Please look for:
 - **ALWAYS** check the previous 2 frames and the following 2 frames for the current label
 - The skeleton is correctly displayed (**NO** missing limbs or weird positions, **NO** missing person in the image) , but allow a leniency of 1-2 frames **IF** the error is small (example: leg cut off in 1 frame - 1, missing person - 0)
(Example: correct - correct - problem- correct -correct => 1 - correct label)
(Example2: correct - correct - problem - problem -correct => 1 - correct label)
 - The skeleton is facing the camera
 - The skeleton can reach the bubbles (no orange lines) , but please allow for a small leniency
(Example: if the skeleton is very close to the only orange lines => 1 - correct)
(Example 2: if the skeleton misses 1-3 bubbles = orange lines, but the lines are **VREY** close => 1 - correct)
(Example 3: if the skeleton misses 1+ bubbles = orange lines **and** the lines are **NOT** close => 0 - correct)
- Please **disregard**: If the skeleton is under the floor